Template updates for audit engagements

Publish once per product and market. Resolve the badge from verified metadata when the auditor lists files. Reopening 20,000 files at 20 concurrent one-minute loads would take 16 h 40 min.

One release, from publication to the auditor's screen / Figure 1

Step	What happens to a Canadian-market release
1. Publish v7	Publish v7, allowed upgrade paths and outbox revision 42 in one shared template DB transaction.
2. Distribute	Regional consumers ignore older revisions and fetch a snapshot on gaps. Prepare common summaries in the background.
3. List 50 files	Read the local catalog and firm index. An auditor who declined v5 for a file on v4 sees the v7 offer. A file on v7 is current.
4. Inspect changes	Resolve the exact hashes, locale and policy. Pin the archived text to a view receipt. On a miss, show "Preparing summary" and enqueue one job for that pair.

Data ownership and tenant isolation

Template service. The shared DB stores all template versions as ZIPs of structured JSON. Lookup by ID and version is fast. Content owners govern immutable releases, parent links, upgrade paths and withdrawals by market. It holds only shared template data.

Engagement manager. On create, it reads the latest applicable template and saves its ID and version inside the file. Its new hook also writes a queryable row in the firm DB (0.8 GB raw at 1 KB/file). The same manager opens files and records decisions. Only confirmed application advances the effective version.

Regional API. Derive the firm and database from the authenticated session. Authorize files on lists and detail views. Exclude exact declines, pin the catalog revision and page by product, track, baseline and file ID. Firm data, receipts, queues, logs, keys and backups stay in-region. Only template data and opaque admission tokens cross regions.

Advantages and disadvantages of the main decisions

Decision	Advantage	Disadvantage
Distribute catalog events	The outbox retains each revision until the relay can deliver it.	While delivery is pending, a regional list may show the old target.
Compute badges on read	One catalog change makes v7 discoverable without writing to every file.	Sparse pending lists may scan many rows. Measure rows scanned per result.
Check decisions with the template authority	Apply and decline requests have a clear order relative to a withdrawal.	A decision adds a service call. An authority outage pauses new decisions.
Limit background verification	The file service retains capacity for live engagements during migration.	Some legacy files wait for verification. At 20 slots, a full backfill takes about 28 days.

The one requirement I would challenge

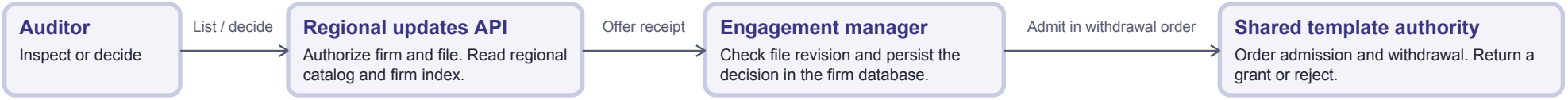
Day-one freshness for every file. "Created from v4" leaves the current version uncertain. Guessing risks showing the wrong changes. Opening the file takes about a minute. I would agree with Product to show updates within seconds for verified files. Until the manager confirms a legacy file's version, show "Needs verification" and withhold its version-specific offer.

Scope. Content application, automatic branch merges and audit conclusions are outside this feature.

End-to-end system architecture

Proposed service boundaries for a firm's home region and shared template services. Firm data, workpapers and receipts stay in the firm's home region.

Offer decisions and release withdrawal



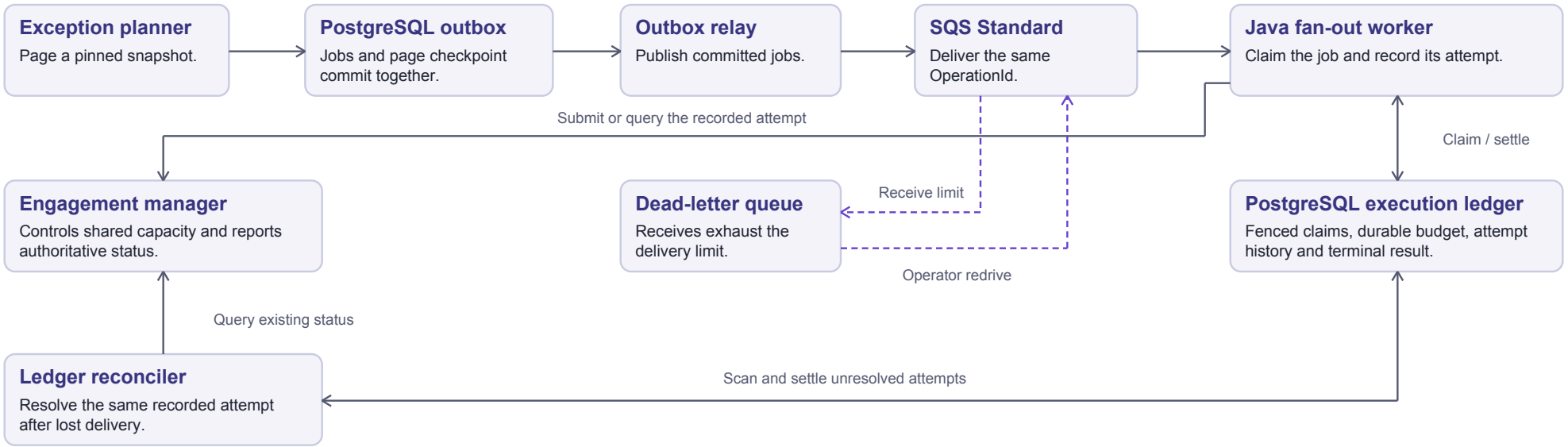
Template publication commits a release and outbox revision together. Regional catalog consumers ignore older revisions and fetch a snapshot on gaps.

Summary lookup and cold miss



Only verified, archived text is shown. The firm database records the exact version and locale shown to the auditor.

Background verification in the same firm region



A terminal ledger commit precedes SQS acknowledgement. The reconciler can settle an existing attempt without a live delivery. Operator redrive preserves the original operation, its history and its remaining budget. See the recovery policy on page 5.

Which update should the auditor see?

Select the content team's active target for the product and market. Require an approved path from the effective baseline. Hold offers with ambiguous targets or missing compatibility data.

Event for one engagement	What the auditor sees or can do
v4 applied, v5 declined	The decline pins this engagement, baseline v4 and immutable target v5. Another file at the same firm can still accept v5.
v6 and v7 accumulate	If v7 is the approved compatible target, show one cumulative summary from v4 to v7. A v5 decline suppresses only that earlier offer.
v7 is withdrawn	Hide v7 from new decisions. A replacement must be designated and compatible. Already-applied v7 remains the baseline. An earlier admission follows the policy below.
Auditor chooses Apply	Record an intent. Show "Application pending" until the owner confirms success or failure. Keep the last confirmed baseline while the outcome is unknown.

Who wins if v7 is withdrawn while it is on screen?

Proposed policy for Product. For each command or retry, the manager checks current decision rights on the file and matches the receipt to the firm, user and file. Bind request ID, payload, file revision and nonce in the firm DB. The authority checks target and epoch, orders admission with withdrawal per track and records the nonce outcome. Withdrawal first: reject and refresh. Admission first: the intent can commit even after withdrawal if its short-lived grant is valid and the file revision matches. Expiry requires a fresh view. Retries reuse the nonce. Return committed results and reject changed payloads. Atomically advance the file revision and store decision, grant, result and outbox. A grant alone cannot change a file.

Security controls across services

Use TLS in transit and regional KMS keys for evidence, queues and logs at rest. Verify encryption of firm databases and backups with their owner. Give each workload a least-privilege IAM role with temporary credentials. The worker calls the owner's API, which checks firm and file scope. The worker cannot read workpaper tables. Log actor, file, action and outcome regionally. Omit payloads, prompts and credentials.

Which part benefits from a model?

Grounding. Figure 2 uses the exact baseline and target. An addition in v5 reversed by v7 drops out of the v4-to-v7 diff. Rules handle simple diffs. In long releases, a model may group related edits across sections. Editors own wording and material exceptions. The model sees no tenant data and has no tools.

Release gate. Editors label whole held-out releases by market and language. Include reversals, wrong topics and polarity flips. Require all critical changes and zero unsupported claims. In a blind crossover trial, ship model grouping only if auditors find changes faster than with rules, with equal or better accuracy and within budget. A valid group can still mislead.

When would a different AI technique earn its cost?

Retrieval. If Product adds methodology explanations, resolve known references by ID. For unlinked guidance, test Bedrock Knowledge Bases with OpenSearch hybrid search over approved passages filtered by market, language and effective date. Require an applicable citation and archive its edition and passage. Otherwise withhold that explanation.

Model changes. Consider fine-tuning with QLoRA on compatible weights only for recurring, labeled grouping errors. Recursive FCoT and voting add calls without verifying facts. Use one bounded call. Revisit agent skills or MCP if several assistants need the same editorial procedure or tools.

From template diff to audit summary

Create one job per template comparison

On publish, warm common pairs. On a regional miss, enqueue once per product, track, base hash, target hash, locale and policy version. The pending badge is already visible while the first summary is prepared.

Ground each claim in the two approved versions

Fetch v4 and v7 by key. Verify both hashes. The existing JSON comparator emits net edits. For each, keep the source path, operation, scope, before/after values and permitted topics.

C17	**/inventory/signoffRequired**	**false to true**	Topics: inventory, review
C18	**/inventory/exceptionReason**	**absent to present**	Topics: inventory, exceptions

Rules summary by default

Typed edits select approved phrases. Use field labels and values from the verified pair. Hold a summary for editor review if a material edit lacks a safe mapping.

When each edit has one topic, group in code and skip inference.

Optional grouping in Amazon Bedrock

For long releases with several valid topics, send only typed facts, allowed topic IDs and a versioned glossary to Nova 2 Lite via Converse. Treat labels as data. Tools are disabled.

Output contract (illustrative group)

```
{"topic":"inventory","ids":["C17","C18"]}
```

What the auditor can read

“Inventory sign-off is now required.” **[C17]**
“Inventory now has a field for the reason for an exception.” **[C18]**

Presence alone says nothing about whether an entry is mandatory.

Validate before promoting a plan

Require valid JSON, every expected ID exactly once and a permitted topic for each ID. Code renders every sentence in the user’s locale. Reject a missing C18, an invented ID or extra prose. On timeout, excess input or budget exhaustion, keep the rules revision.

Preserve the summary the auditor actually saw

Store exact text, sentence links, diff, hashes, glossary, policy, prompt, model ID and checks in versioned S3 under Object Lock for the agreed retention period. Verify the regional copy before promotion. A late rules job cannot replace a promoted revision.

The regional API records the artifact version, file revision, catalog epoch, user, locale and time in the firm database before returning the text. A November review retrieves the March artifact and traces its sentences back to C17 and C18.

Evolution and recovery in production

Proposed regional operations. The local exercise verifies PostgreSQL polling and controlled HTTP dependencies.

Roll out without a maintenance window

Install tables, outboxes and create, open and decision hooks before the inventory watermark. Capture metadata on opens and backfill resumably (about 60 s per file). Exclude volatile values. Check baselines and declines against manager history, flag unknowns and reject stale repairs. Exercise cross-firm isolation and read-only access before launch. Disable affected routes on an isolation failure and investigate.

Expand, migrate and contract

Expand. If pending-only pages exceed 500 ms p95 for five minutes and query plans show sparse scans, add a regional index from verified metadata. Keep authorization on every read. Deploy consumers that understand both event versions before switching producers.
Migrate. Shadow the query in a small and a large firm per region. Compare IDs and offers at matching revisions, including declines and withdrawals. Enable by firm flag if results match, p95 stays below 500 ms and projected cost is below X. Fall back to live reads on index lag.
Contract. Retire obsolete readers after the rollback window and successful replay. Preserve archived summaries and receipts.

Failure policy by dependency

Worker crash or status timeout	Recover the original operation and attempt from the ledger. A missing status leaves the outcome uncertain. The manager keeps capacity until confirmed completion or enforced cancellation. Trial deadlines: 3 s connect, 120 s server, 125 s total.
New attempt or terminal result	A replacement submission requires confirmed no effect, a fence against late execution and a remaining durable authorization. Commit success, permanent rejection or confirmed budget exhaustion as terminal, then delete the SQS delivery. A duplicate terminal delivery reuses that result.
Dead-letter delivery and redrive	SQS transfers repeatedly received unacknowledged deliveries at the configured receive limit. The ledger reconciler still queries uncertain attempts. Diagnose and redrive the message with its original operation ID. History, result and remaining budget stay in the ledger. Redrive creates no new authorization.
Catalog or grouping outage	Pause new admissions and label stale reads. Use archived verified rules if grouping is unavailable. Initial Bedrock target: 15 s per attempt and one budgeted retry on 429, 503 or timeout. An unknown apply stays pending until the manager confirms its outcome. Only confirmed success advances the baseline.

Failover, capacity and service objectives

Run API replicas in two availability zones. Replace failed tasks, confirm database standby promotion and throttle on a zone loss. For region loss, pause affected firms, restore in-region and reconcile receipts against agreed RTO and RPO. Bound worker concurrency under the manager's assumed 20 shared slots. Trial budget: three downstream submissions per operation. Apply backoff with jitter and respect Retry-After on transient failures. Pause backfill when the database saturates. Provisional regional goals are 99% of catalog revisions visible within 10 s and p95 below 500 ms for 50 files. Monitor queue age, unresolved attempts and verification lag. Drill zone loss, redelivery, lost replies, withdrawals and grouping outages.

Delivery and redrive rules Regional readiness tests

Scaling limits and monthly cost

Which limit binds first?

Workload	Proposed operating point and what to validate
Releases and reads	40 × 52/12 ≈ 173 releases/month. Mean product: 20,000 files. At 50 files/page, the median firm fits on one page and the largest needs 800 for a full scan. List pages: 4k firms × 10/day × 20 days = 0.8M. Polls: 100 tabs × 8 h × 20 days × 720/h = 11.52M. Test sparse lists alongside small firms.
Interactive capacity	Assume a mixed 60 requests/s peak in one region. Target 15/s per replica only if a canary sustains 25/s at p95 ≤ 500 ms on the priced task size. Four serving replicas, five for one failure, six during rollout. Eight connections across all pools per replica means up to 48 list sessions in that region, plus write capacity. Load-test the mix and connection waits.
Slow file opens	20 owner slots × 60 s/min ÷ 60 s/load = 20 loads/minute. For 800,000 unknown files: 13,333 call-hours, about 27.8 days at full utilization or 55.6 days at 50%. Extra workers cannot raise the owner's limit.

Monthly scenario in USD / Frankfurt and Canada Central

On-demand list rates checked 10 Nov 2026. Assume Linux/x86 ECS Fargate, 730 h/month, two regional deployments and gross paid usage before tax, credits or discounts. Per-region volumes below are sizing assumptions.

Cost driver	Monthly units and assumptions	USD
ECS Fargate	2 API tasks/region (1 vCPU, 2 GB) $162.16. Frankfurt peak 120 task-h + 1 rollout task-h/region $6.92. One 0.5 vCPU, 1 GB worker/region $40.54.	209.63
ALB, IPv4, transfer	1 ALB and 1 average LCU/region $50.04. Four public IPv4 addresses $14.60. Per region: 25 GB internet out + 1 GB template transfer to the other region $4.54 combined.	69.18
S3, SQS, KMS	Per region: 25 GB S3 Standard, 5k PUT and 50k GET ($1.34 combined). 100k SQS units ($0.08). One KMS key and 50k calls ($2.30).	3.72
CloudWatch	Per region: 10 GB log ingestion, 5 GB-month archived, 10 custom metrics and 5 standard alarms.	19.13
Bedrock Nova 2 Lite	Global profile via US East, template facts only. 173.33 × 4 baselines × 2 locales × 1.25 attempts × (3k input at $0.30/M + 500 output at $2.50/M).	3.73
AWS usage subtotal	Sum at full precision. List traffic and polling run on the priced API and ALB.	305.38
Internal assumptions	Shared SQL, backups, platform and network allocation $150. 8k owner loads × $0.01 = $80. Operational review 2 h × $60 = $120.	350.00
Monthly envelope	**($305.3781 + $350) × 1.20 = $786.4537. Reserve $787/month for planning.**	**786.45**

Budget gate. Confirm X, owner cost/load and shared allocations with Finance. Full migration adds $8,000 before retries if $0.01/load holds. Each $0.01 increase per steady-state load adds $96/month including reserve. If each region needs a dedicated Multi-AZ db.t4g.medium PostgreSQL deployment with 20 GB gp3, add $222.26/month before reserve. Pace migration from headroom only while projected spend stays below X.

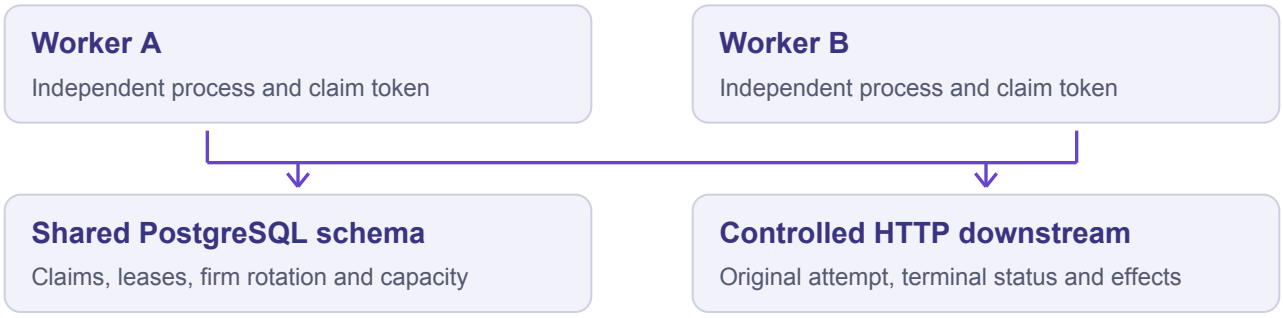
Rate evidence. See the companion cost workbook for units, regional rate cards and an editable X. AWS Price List Bedrock pricing

Distributed worker tests

MAIN 362EF1C / JAVA 25 / SPRING BOOT 4.1.2 / POSTGRESQL 18.6

Pages 1-6 describe the proposed regional deployment. This appendix records behavior verified at the baseline commit shown above.

Two independent JVMs in the worker process tests



Both JVMs use the same PostgreSQL schema and HTTP fixture. The harness checks durable state, effects per operation and peak downstream concurrency.

Scenario	Verified behavior
Process loss	The survivor reclaims the original attempt and reconciles its outcome. Each operation applies once and the measured peak is one.
Late reply, same token	The lease expires before the reply. The current token and attempt still match, so the first JVM settles the confirmed result and frees the slot.
Late reply, replaced token	A second JVM reclaims the attempt. The first JVM's update affects zero rows. The reservation stays active until the second JVM settles the same attempt.
Capacity and firm rotation	At capacity one, the next job waits for settlement. At capacity two, the small firm takes the next slot while 30 large-firm jobs remain ready.
Snapshot-based planning	A fresh planner JVM resumes with the pinned snapshot and cursor after the source changes. Expiration or a mismatched snapshot halts planning.

In both late-reply tests, the next job advances, each operation applies once and peak concurrency is one.

With PostgreSQL running, use **./mvnw -Ppostgres-integration -Dit.test=DurableWorkerProcessIT,PublicationSnapshotIT verify**. The separate smoke test runs the one-JVM demo.

48 unit and 31 PostgreSQL integration tests passed. Worker test reports JaCoCo and SonarQube reports

Version and summary validation

MAIN 362EF1C / REFERENCE APIS OVER LOOPBACK HTTP

The reference APIs use controlled fixtures. The connected suite uses a test driver to pass the offered version pair between two local Spring Boot servers.

What the HTTP suites verify

HTTP suite	Verified behavior
Version selection **21 scenarios**	Offer and decision rules cover stale revisions, catalog withdrawal, firm isolation and confirmed apply outcomes. A stale receipt returns HTTP 409. Replaying apply preserves the decision ID and replaying decline returns HTTP 204.
Summary validation **15 scenarios**	Approved SHA-256 hashes and the exact pair determine the typed diff. The suite verifies C17 and C18, grouping rejection, rules fallback, editorial review and localized receipt retrieval.
Connected references **3 scenarios**	The driver forwards the offered pair and preserves the original French Canadian v7 receipt after v8. A withheld offer skips summary generation. Tampered bytes return HTTP 422.

Two receipts

Receipt	Purpose
Offer receipt	The regional reference uses an opaque receiptId to bind an apply or decline decision to the original offer, firm, file, engagement revision and catalog epoch.
Summary view receipt	The summary reference pins the version, locale, approved hashes, text and change IDs shown to the auditor.

CI and regional rollout

118 tests passed on main. The required **verify** check waits for worker quality and all three HTTP suites. JaCoCo reported full line and branch coverage. SonarQube passed with 100% worker coverage. Both run on pull requests.

The fixtures verify the local reference contracts. Regional rollout requires evidence for deployed services, durable storage and SQS delivery.

Version selection Summary validation Connected flow Main CI run Evidence index

Firm scheduling during reconciliation

DISTRIBUTED WORKER / TWO JVMs / SHARED POSTGRESQL

Slow status lookups for unresolved attempts can consume every worker's polling time. Each worker now alternates its first choice between due status lookups and new job admissions when both can proceed.

How each polling loop chooses work

Resolve a due attempt
Keep the original attempt and its existing reservation.

→

Try a fresh admission
Use the next eligible firm's turn if a slot is free.

→

Check due status again
A fresh claim restores status lookup as the first choice.

Workers start by checking for a due status lookup. The last successful claim determines the next preference. If the preferred class has no eligible claim, the worker tries the other. PostgreSQL keeps the capacity check, new reservation and firm cursor update in the same transaction.

Two-JVM regression

Checkpoint	Assertion
Starting state	At global capacity five, four large-firm attempts are SUBMITTED. The mock holds two PENDING status replies, two other lookups are due and a small-firm job becomes READY.
Progress	Release one held status reply. The small firm's job reaches APPLIED while all four original large-firm attempts remain unresolved and keep their reservations.
Shared limit and effects	The small firm's operation applies exactly once. The measured peak is five, equal to the configured global limit.

Reproduce the scenario

With PostgreSQL running, use **./mvnw -Ppostgres-integration -Dit.test=DurableWorkerProcessIT verify**. The test starts two independent worker JVMs against one PostgreSQL schema and a controlled local HTTP downstream.

Worker scheduling design Scenario evidence Latest main CI runs Wiki architecture